

Operational Enumeration in Rational Pushdown Normalization

Schubert Cancellations, Raney Carry Histories, and One-Counter Compression

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Abstract

We enumerate the physical events produced by a centered real-time deterministic pushdown normalization process for the rational balance language

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p,q) = 1.$$

The work originates from a concrete completion-aware 2:1 pushdown automaton and follows the same residual semantics to arbitrary coprime $p:q$. Rather than counting only accepted words, we count actual MATCH/CANCEL and physical WRITE events along rational-Dyck inputs.

In the primitive sector, no WRITE can occur after the first symbol. Consequently the cancellation statistic is a fixed-stack Hamming agreement statistic. We obtain its full balanced distribution and, on primitive rational-Dyck paths, identify it with a basis-intersection statistic of an explicit Schubert/lattice-path matroid. Ordinary and rational Narayana distributions appear in the near-diagonal cases, while Lorentzian matroid theory yields ultra-log-concavity of the general normalized primitive coefficients.

For repeated kernels we refine the classical Bizley–Duchon factorization by CANCEL and OUTER-WRITE. We prove an exact carry-onset hierarchy, a pathwise WRITE-weight conservation law, and a Raney law for minimal multi-carry signatures. Between successive carries the cancellation statistic is governed by a stationary shifted-Schubert kernel whose bulk generating function is rational; the complete minimal-onset family is therefore described by a finite algebraic small-root/boundary system. Finally, a centered phase-normal-form theorem reconstructs every rational-chamber stack from one residual counter and one phase bit, rules out negative OUTER writes, and reduces arbitrary-scale event enumeration to an exact weighted one-counter recurrence. The original 2:1 family then admits the all-scale identity $C(P) = k + s(P)$, where $s(P)$ is the number of zero-residual factors.

The Narayana, Schubert-matroid, Lorentzian, Bizley–Duchon, Raney, and kernel-method ingredients used after these operational identifications are classical. The contribution is the normalization-specific event theory and the exact bridges by which the physical pushdown dynamics enters those structures.

Keywords. deterministic pushdown automata; rational Dyck paths; operational enumeration; cancellation statistic; Schubert matroid; Narayana numbers; Raney numbers; Bizley factorization; kernel method; one-counter automata.

1 Introduction

A pushdown normalizer does more than accept or reject a word. It performs a history of physical actions: it writes a shortest completion demand, searches through that demand when the next input disagrees with the visible top symbol, cancels cells when input and demand agree,

and occasionally performs a larger outer reset. These events depend on the chosen real-time normalization schedule even when the accepted language is unchanged. The purpose of this paper is to enumerate those actual events for one canonical centered schedule and to identify the intrinsic combinatorial structures hidden by the operational description.

The discovery route is important. The project began with the explicit completion-aware 2:1 automaton in Figure 1, designed for $N_1 = 2N_0$. Its center/left/right control distributes an unresolved completion demand between finite control and a binary stack. The scalar residual

$$\rho_{2,1} = N_1 - 2N_0$$

extracted from that machine generalizes immediately to

$$\boxed{\rho_{p,q} = qN_1 - pN_0.}$$

The centered $p:q$ normalizer then turns this residual into a canonical shortest nonnegative demand. Once the general machine was available, a new question became natural: what do its *physical* CANCEL and WRITE histories count on the rational-Dyck chamber $\rho_{p,q} \geq 0$?

This paper answers that question from primitive paths through arbitrary scale. The logical development is

$$\boxed{\begin{array}{l} 2:1 \text{ DPDA} \longrightarrow \rho_{p,q} \longrightarrow \text{centered } p:q \text{ normalization} \\ \longrightarrow \text{CANCEL/WRITE event enumeration} \end{array}}$$

The same 2:1 figure is retained because it records the concrete origin of the residual and completion semantics; none of the later enumerative proofs depends on treating that special machine as a hypothesis.

1.1 The motivating 2:1 automaton

A transition label $a, A/\gamma$ consumes input a and replaces the top stack symbol A by γ ; the leftmost work symbol is the stack top. The bottom marker is Z_0 . Dashed boxes contain right-endmarker completion operations, not ordinary ε -moves during binary-input processing.

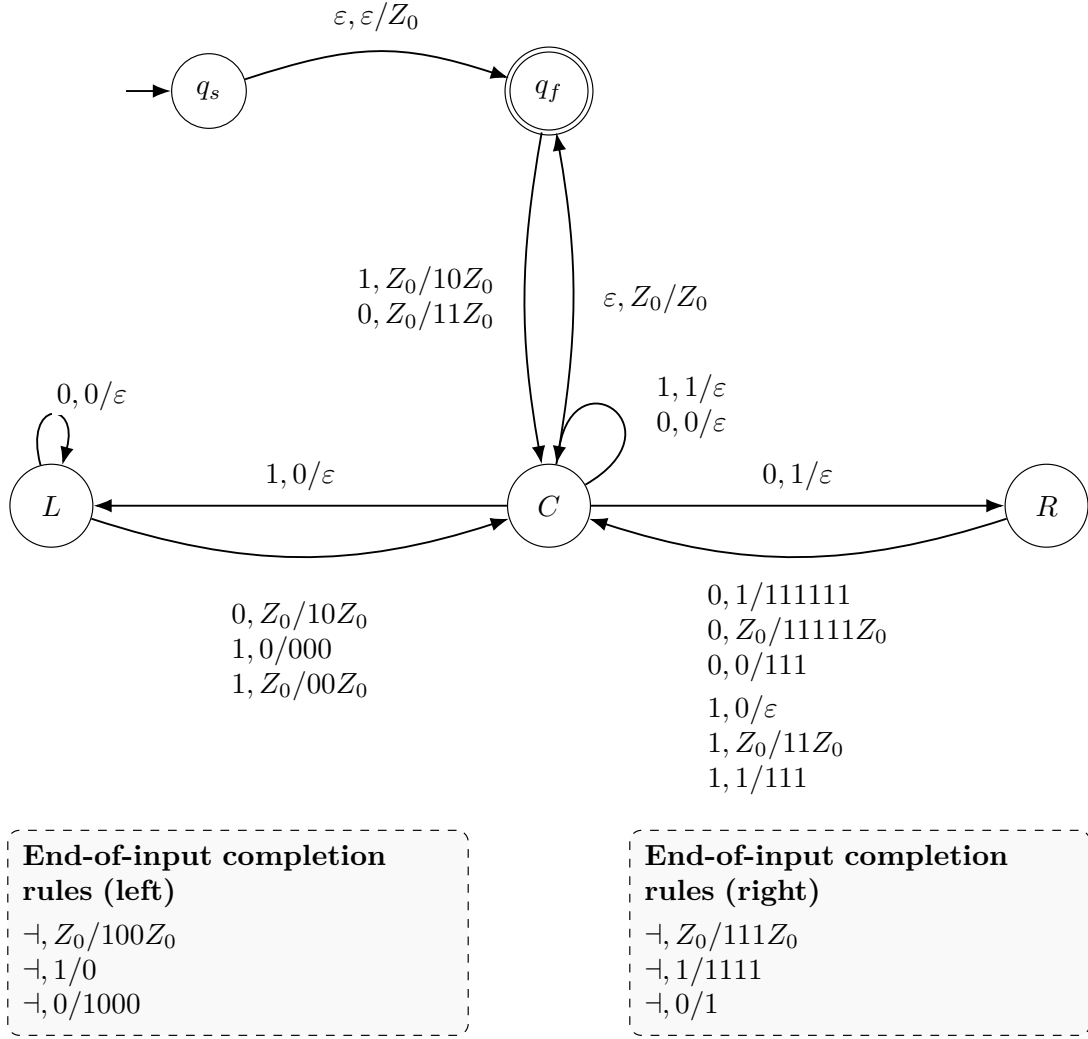


Figure 1: The completion-aware 2:1 DPDA that motivated the residual and shortest-demand viewpoint. The present paper does not enumerate this historical schedule directly; it uses the centered $p:q$ schedule defined below and returns to 2:1 as an exact specialization.

1.2 Main results and novelty boundary

The main results are operational identifications, not new definitions of the classical objects that appear after the identification.

1. Primitive balanced normalization is a fixed-stack Hamming/Johnson enumerator, with an explicit complete CANCEL distribution.
2. Primitive rational-Dyck CANCEL becomes a basis-intersection statistic in a Schubert/lattice-path matroid; ultra-log-concavity follows from Lorentzian matroid theory.
3. Zero-residual factorization lifts Bizley–Duchon structure to a bivariate CANCEL/OUTER-WRITE generating system.
4. Physical carries obey an exact onset hierarchy $\kappa_r = a + rc$, and the scale is the total physical WRITE weight.
5. Minimal multi-carry signatures are Raney objects; their edges carry stationary shifted-Schubert cancellation polynomials with a rational bulk kernel.
6. Every rational-chamber stack satisfies a centered phase normal form. This excludes OUTER– and compresses the arbitrary-scale machine to finite control plus one residual counter.
7. In the original 2:1 family, arbitrary-scale CANCEL has the pathwise form $C(P) = k + s(P)$.

Rational Narayana numbers, lattice-path matroids, Lorentzian theory, Bizley–Duchon factorization, Raney numbers, and small-root kernel methods are classical and are cited as such below [2, 6, 7, 5, 8, 4, 10, 3].

2 The centered $p:q$ event process

Fix coprime positive integers p, q , put

$$m = p + q, \quad h = \left\lfloor \frac{m-1}{2} \right\rfloor,$$

and write D_d for the centered debt state with $-h \leq d \leq h$. For an integer residual r , let

$$\mu(r) = (x, y) = \arg \min \{x + y : x, y \in \mathbb{Z}_{\geq 0}, px - qy = r\},$$

which is unique after restricting to the standard nonnegative kernel strip. The canonical binary realization of a demand pair is

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y. \end{cases}$$

The centered real-time process has four event classes. If the stack top matches the next input bit, the top cell is popped (MATCH/CANCEL). A mismatch is absorbed by changing d by one while $|d| < h$ (SEARCH $\pm$). At the outer boundary a further same-direction mismatch performs OUTER-WRITE and returns to debt 0. If the stack is empty, the next symbol performs BOTTOM-WRITE using the shortest-demand map μ .

The exact transition rule is the following. With input bit a and stack word S whose leftmost symbol is the top,

$$\begin{aligned} \text{top } z = a : & & (d, zT) &\mapsto (d, T), \\ (a, z) = (1, 0), d < h : & & (d, 0T) &\mapsto (d+1, T), \\ (a, z) = (0, 1), d > -h : & & (d, 1T) &\mapsto (d-1, T), \\ (a, z) = (1, 0), d = h : & & (h, 0T) &\mapsto (0, \eta(\mu((h+1)m))T), \\ (a, z) = (0, 1), d = -h : & & (-h, 1T) &\mapsto (0, \eta(\mu(-(h+1)m))T), \\ S = \varepsilon : & & (d, \varepsilon) &\mapsto (0, \eta(\mu(dm + w(a)))) \end{aligned}$$

where $w(1) = q$ and $w(0) = -p$. The language-recognition correctness of this centered normalization is background for the present event-enumeration problem; all statistics below refer to these displayed transitions.

A word is *rational-Dyck* when every prefix has

$$\rho = qN_1 - pN_0 \geq 0.$$

A scale- k balanced word has kp ones and kq zeros. Let $C(P)$ be its number of MATCH/CANCEL events and $O(P)$ its number of OUTER-WRITE events.

3 Primitive balanced words: fixed-stack Hamming geometry

A primitive balanced word has p ones and q zeros.

Lemma 3.1 (No later WRITE in the primitive sector). *After the first input symbol of a primitive balanced word, the shortest-demand stack has length $m-1$. Exactly $m-1$ input symbols remain. Hence no later WRITE occurs: every remaining transition is SEARCH or CANCEL.*

Proof. The first BOTTOM-WRITE realizes the unique shortest completion of the first-symbol residual. Because the completed word has total composition (p, q) , the demand length is $m - 1$. Every subsequent non-WRITE transition consumes one input symbol and one stack cell. A new WRITE before the end would create more stack cells than remaining input can remove, contradicting balanced acceptance. $\square$

Thus $C(P)$ is exactly Hamming agreement between the suffix of P and its fixed canonical stack reference.

Theorem 3.2 (Primitive balanced CANCEL distribution). *For every coprime $p, q > 0$ with $p \neq q$,*

$$c_{\min}(p, q) = |p - q| - 1.$$

The number of primitive balanced words attaining this minimum is

$$\binom{\max(p, q) - 1}{|p - q| - 1} = \binom{\max(p, q) - 1}{\min(p, q)}.$$

In the unique coprime diagonal case $(p, q) = (1, 1)$, both primitive balanced words 10 and 01 have exactly one CANCEL, so

$$c_{\min}(1, 1) = 1$$

with multiplicity 2. More generally, for $j \geq 0$,

$$\#\{P : C(P) = m - 1 - 2j\} = \binom{p-1}{j} \binom{q}{j} + \binom{p}{j} \binom{q-1}{j},$$

with all other values zero.

Proof. Condition on the first bit. Lemma 3.1 reduces the suffix to comparison of two binary strings of equal length and fixed composition. If the first bit is 1, the input suffix contains $p - 1$ ones and q zeros; if it is 0, it contains p ones and $q - 1$ zeros. A Hamming disagreement swaps one selected 1-position with one selected 0-position, so disagreements occur in pairs. Choosing j swapped positions in the two composition classes gives the two binomial products. For $p \neq q$, maximizing the number of paired disagreements gives $|p - q| - 1$ agreements and the stated multiplicity. In the remaining coprime case $(1, 1)$ the distribution formula itself gives two words at CANCEL value 1, yielding the separate diagonal statement. $\square$

Corollary 3.3 (Cancellation-free primitive orbit). *A cancellation-free primitive balanced word exists if and only if $|p - q| = 1$. It is unique: for $p = q + 1$ it is $(10)^q 1$, while for $q = p + 1$ it is $0(01)^p$.*

This is the first operational appearance of the Christoffel/mechanical-word boundary: the extremal word is a cyclic representative of the corresponding balanced mechanical pattern, but no Christoffel theorem is needed in the proof.

4 Primitive rational-Dyck paths and the Schubert statistic

Assume first $p = q + d$ with $d \geq 1$; the opposite inequality follows by the exact bit/physical duality. Define

$$R_{q,d} = 1(10)^q 1^{d-1}, \quad A_q = \{3, 5, \dots, 2q + 1\}.$$

For a primitive rational-Dyck path P , let $Z(P)$ be the set of zero positions.

Theorem 4.1 (Canonical-demand agreement theorem). *For every primitive rational-Dyck P of type $(q + d) : q$,*

$$C(P) = d - 1 + 2|Z(P) \cap A_q|.$$

Moreover, writing $Z(P) = \{z_1 < \dots < z_q\}$, rational-Dyck nonnegativity is equivalent to

$$z_i \geq \left\lceil \frac{(2q + d)i}{q} \right\rceil, \quad 1 \leq i \leq q.$$

Hence the zero sets form the bases of a Schubert, equivalently lattice-path, matroid.

Proof. The first BOTTOM-WRITE fixes the entire remaining demand by Lemma 3.1. Direct comparison with that demand yields the reference word $R_{q,d}$ and therefore the displayed agreement formula. At the position z_i of the i th zero, the prefix has $z_i - i$ ones and i zeros; $q(z_i - i) - pi \geq 0$ is exactly the stated lower flag. Such coordinatewise lower flags define the standard Schubert/lattice-path basis family [6]. $\square$

Define the normalized polynomial

$$\mathcal{R}_{q,d}(x) = \sum_B x^{|B \cap A_q|},$$

where B runs over these bases.

Corollary 4.2 (Ultra-log-concavity). *If $\mathcal{R}_{q,d}(x) = \sum_j r_j x^j$, then*

$$\left(\frac{r_j}{\binom{q}{j}} \right)^2 \geq \frac{r_{j-1}}{\binom{q}{j-1}} \frac{r_{j+1}}{\binom{q}{j+1}}.$$

Thus the primitive normalized CANCEL coefficients are ultra-log-concave, and hence log-concave and unimodal with no internal zero.

Proof. The multivariate basis polynomial of a matroid is Lorentzian; nonnegative specialization to a marked subset preserves the required Lorentzian consequences. Apply the strongest Mason inequality of Brändén–Huh [7]. $\square$

The support is exact:

$$j_{\max} = \left\lfloor \frac{q(2q + 1)}{2q + d} \right\rfloor, \quad j_{\min} = \begin{cases} 1, & d = 1, \\ 0, & d \geq 2. \end{cases}$$

After reversal to an upper flag, put

$$s_i = 1 + \left\lfloor \frac{(2q + d)(i - 1)}{q} \right\rfloor, \quad \bar{A}_q = \{d, d + 2, \dots, d + 2q - 2\}.$$

If $u_i = |\bar{A}_q \cap [1, s_i]|$, $v_i = s_i - u_i$, and

$$\sum_{r \geq 0} H_{i,r}(x) z^r = (1 - xz)^{-u_i} (1 - z)^{-v_i},$$

then the standard flagged Jacobi–Trudi/Lindström–Gessel–Viennot specialization [11, 12] gives

$$\mathcal{R}_{q,d}(x) = \det[H_{i,1-i+j}(x)]_{1 \leq i,j \leq q}.$$

Because the determinant is upper Hessenberg it also has the matrix-free recurrence

$$D_{q+1} = 1, \quad D_i = \sum_{r=1}^{q-i+1} (-1)^{r-1} H_{i,r} D_{i+r}, \quad \mathcal{R}_{q,d} = D_1.$$

4.1 Narayana windows

The near-diagonal cases recover classical distributions after the operational identification. If $d = 1$, deleting the initial 1 gives an ordinary Dyck path of semilength q , and

$$\#\{P : C(P) = 2j\} = \frac{1}{q} \binom{q}{j} \binom{q}{j-1}.$$

If $d = 2$ (so q is odd by coprimality), a last-level decomposition $P = U A U B$ gives

$$\boxed{\#\{P : C(P) = 2j - 1\} = \frac{1}{j} \binom{q+1}{j-1} \binom{q-1}{j-1}.}$$

These are ordinary/rational Narayana distributions; the second equality is distributional, not pathwise equality with the classical peak statistic [2].

5 Repeated kernels and operational Bizley factorization

For scale k , let $\mathcal{D}_k$ be the rational-Dyck paths with kp ones and kq zeros, and let $\mathcal{I}_k$ be those with no proper zero-residual contact. Put

$$D_k(x, y) = \sum_{P \in \mathcal{D}_k} x^{C(P)} y^{O(P)}, \quad I_k(x, y) = \sum_{P \in \mathcal{I}_k} x^{C(P)} y^{O(P)}.$$

A zero-residual contact returns the centered machine to the canonical accepting configuration. Both statistics are additive under concatenation, hence

$$\boxed{D(z; x, y) = 1 + \sum_{k \geq 1} D_k(x, y) z^k = \frac{1}{1 - I(z; x, y)}.$$

Define operational Bizley cumulants by

$$\boxed{\log D(z; x, y) = \sum_{k \geq 1} \frac{A_k(x, y)}{k} z^k.}$$

At $x = y = 1$ this reduces to the classical rational-Dyck/Bizley decomposition; in particular

$$\boxed{A_k(1, 1) = \frac{1}{p+q} \binom{k(p+q)}{kp}.$$

The unweighted factor-free structure is classical [5, 8, 4]; the refinement here is by actual normalization events.

6 Physical carry hierarchy and WRITE-weight conservation

Set

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \quad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

A positive OUTER-WRITE is called a *carry*.

Theorem 6.1 (Carry-onset hierarchy). *The first scale at which $r \geq 1$ carries are possible is*

$$\kappa_r(p, q) = a + rc.$$

Equivalently,

$$O_{\max}(k) = \max \left\{ 0, \left\lfloor \frac{k-a}{c} \right\rfloor \right\}.$$

At $k = \kappa_r$, every path realizing r carries is factor-free, so

$$[y^r]A_{\kappa_r}(x, y) = \kappa_r[y^r]D_{\kappa_r}(x, y).$$

Proof. Before the first positive outer carry, SEARCH+ must accumulate enough positive centered debt to cross the outer boundary. Among rational-Dyck prefixes with a fixed number of completed scale blocks, replacing a zero by a one cannot delay that positive boundary: a zero lowers the rational height and never supplies positive mismatch budget. Hence the all-ones trajectory is extremal for earliest carry onset. Writing $h = q(a-1) + s$ with $0 \leq s < q$ shows that $a = 1 + \lfloor h/q \rfloor$ scale units are necessary and sufficient before the first OUTER+ trigger.

After an OUTER+ reset the written frame has length $cm - 1$, where $c = 1 + \lfloor h/p \rfloor$. The WRITE-weight invariant of Theorem 6.2 assigns this reset weight c . Between two consecutive carries no event can create scale budget; SEARCH and CANCEL only consume existing cells. Thus each further carry costs at least an additional c units of total scale, giving the lower bound $k \geq a + rc$ for r carries. The same extremal positive-boundary trajectory, followed at each stage by its shortest balancing continuation, realizes equality, so $\kappa_r = a + rc$ and the displayed formula for $O_{\max}(k)$ follows.

At $k = \kappa_r$, suppose a path with r carries had a proper zero-residual factorization into positive scales $k_1 + \dots + k_s = k$, $s \geq 2$, with carry counts $r_1 + \dots + r_s = r$. Every factor with $r_i > 0$ satisfies $k_i \geq a + r_i c$, while a factor with $r_i = 0$ still has $k_i \geq 1$. If carries occur in two or more factors, summing the onset bounds contributes at least one extra copy of a ; if all carries occur in one factor, every remaining factor contributes at least one extra unit of scale. In either case $k > a + rc = \kappa_r$, a contradiction. Hence every onset path with r carries is factor-free. The standard logarithmic relation between all paths and factor-free factors then gives $[y^r]A_{\kappa_r} = \kappa_r[y^r]D_{\kappa_r}$. $\square$

The hierarchy is a consequence of a stronger pathwise invariant.

Theorem 6.2 (WRITE-weight conservation). *Assign every WRITE event a positive integer scale weight ω : a written frame of length $\omega m - 1$ at an empty stack has weight ω , while OUTER+ has fixed weight c and OUTER− fixed weight a . For a balanced accepted scale- k computation,*

$$\sum_{\text{WRITE } j} \omega_j = k.$$

Proof. Every WRITE of weight ω changes physical stack height by $\omega m - 1$, while every SEARCH or CANCEL decreases height by one. If W is the number of WRITES and N the number of non-WRITES, empty-to-empty height balance gives

$$N = m \sum_j \omega_j - W.$$

Since every input symbol produces exactly one event, $W + N = km$, and the identity follows. $\square$

For BOTTOM-WRITE at debt d the exact weight is

$$\omega_B(d, 1) = \max \left\{ 1, \left\lceil -\frac{d}{q} \right\rceil, \left\lceil \frac{d+1}{p} \right\rceil \right\},$$

$$\omega_B(d, 0) = \max \left\{ 1, \left\lceil \frac{1-d}{q} \right\rceil, \left\lceil \frac{d}{p} \right\rceil \right\}.$$

Thus scale is literally the total physical shortest-demand WRITE budget.

7 Minimal multi-carry paths: Raney signatures and Schubert edge weights

Define the first-carry reserve

$$\boxed{R_0 = aq - h}, \quad Q = qc.$$

At minimal r -carry onset let z_j be the number of zeros read before the j th carry. The feasible signatures are exactly

$$0 \leq z_1 \leq \cdots \leq z_r, \quad z_j \leq R_0 - 1 + (j-1)Q.$$

Introduce the zero-tail slack

$$\lambda_j = R_0 - 1 + (j-1)Q - z_j.$$

Then

$$\boxed{0 \leq \lambda_1 \leq R_0 - 1, \quad 0 \leq \lambda_j \leq \lambda_{j-1} + Q.}$$

Theorem 7.1 (Raney signature law). *The number of feasible r -carry signatures is*

$$\boxed{|\mathcal{Z}_r| = \frac{R_0}{(Q+1)r + R_0} \binom{(Q+1)r + R_0}{r}}.$$

Hence the signature skeleton is the Raney number $R_r^{(Q+1, R_0)}$.

Proof. Interpret z_j as the number of horizontal steps before the j th vertical step. The displayed bounds are exactly the strict ballot boundary $x < R_0 + Qy$. The classical Raney/cycle lemma gives the formula [10]. $\square$

Before a carry with slack λ , the state is $(D_h, 0^{\lambda+1})$. After the carry, define

$$X = h + 1 + Q, \quad Y = pc - h - 1.$$

The written frame is

$$W_+ = (10)^Y 0^{X-Y}.$$

For a transition $\lambda \rightarrow \mu$ put

$$\delta = Q + \lambda - \mu, \quad n = pc - 1 + \delta.$$

The no-WRITE phase contains $pc - 1$ ones and δ zeros and is compared with

$$Q_{\lambda, \mu} = (10)^Y 0^{n-2Y}.$$

If $B = \{b_1 < \cdots < b_\delta\}$ is the input-zero set and

$$H_\lambda = (h+1)m + p\lambda,$$

then rational nonnegativity is the lower flag

$$\boxed{b_i \geq \ell_i(\lambda) = \max \left\{ i, \left\lceil \frac{mi - H_\lambda}{q} \right\rceil \right\}}.$$

Let $A_{\lambda,\mu}$ be the zero-position set of $Q_{\lambda,\mu}$. The inter-carry polynomial is

$$M_{\lambda,\mu}(x) = \sum_B x^{n-|A_{\lambda,\mu}|-\delta+2|A_{\lambda,\mu}\cap B|}.$$

Thus each edge of the Raney signature skeleton is a shifted-Schubert Hamming enumerator.

After reversal, $M_{\lambda,\mu}$ has the standard flagged determinant and upper-Hessenberg recurrence. This gives an exact finite product/sum over signatures for every minimal-onset polynomial $G_{p,q}^{(r)}(x)$.

7.1 Stationary bulk kernel and finite wall

When the rational lower flag is inactive, $M_{\lambda,\mu}$ depends only on $\delta = Q + \lambda - \mu$:

$$\widehat{M}_\delta(x) = \sum_{j=\max(0,\delta-Y)}^{\delta} \binom{h+\delta}{j} \binom{Y}{\delta-j} x^{Y-\delta+2j}.$$

Its generating function is the rational symbol

$$\widehat{K}(v; x) = \sum_{\delta \geq 0} \widehat{M}_\delta(x) v^\delta = \frac{(x + (1-x^2)v)^Y}{(1-xv)^{pc}}.$$

For every $p > q$ the wall is absent. For steep $q > p$, let

$$\Theta = pQ - m(h+1), \quad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

If $\Theta > 0$, then

$$b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor, \quad 1 \leq b \leq Q-1,$$

and only target slacks $0, \dots, b-1$ differ from the Toeplitz bulk. The exact operator is therefore Toeplitz-Hessenberg bulk plus a finite-rank rational-wall correction.

8 Algebraic closure of the minimal-onset family

The bulk kernel equation is

$$v^Q = t\widehat{K}(v; x),$$

or

$$v^Q(1-xv)^{pc} = t(x + (1-x^2)v)^Y.$$

Let $v_1, \dots, v_Q$ be the Q small Puiseux branches, taken in an algebraic Puiseux extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric Green-function expressions below descend to algebraic formal series over $\mathbb{Q}(x, t)$; see Appendix C. The bulk excursion Green function is

$$\mathcal{E}_Q(t; x) = \frac{(-1)^{Q-1}}{tx^Y} \prod_{i=1}^Q v_i,$$

and from slack 0 to slack μ ,

$$\mathcal{R}_\mu(t; x) = \mathcal{E}_Q(t; x) h_\mu(v_1, \dots, v_Q),$$

where h_μ is the complete homogeneous symmetric polynomial. These are the standard small-root half-line formulas specialized to the DPDA-derived symbol [3].

For a general start $s < Q$, the Green row has the finite form

$$\mathcal{R}_\mu^{(s)} = \sum_{i=1}^Q c_i^{(s)} v_i^\mu.$$

The coefficients are determined by a $Q \times Q$ boundary matrix. Writing

$$K_{<n}(v; x) = \sum_{d=0}^{n-1} \widehat{M}_d(x) v^d, \quad \beta_n(v) = tv^{-n} K_{<n}(v; x),$$

define

$$C_{\mu i} = \beta_{Q-\mu}(v_i), \quad 0 \leq \mu < Q.$$

Then

$$\boxed{C C^{(s)} = e_s.}$$

The first-carry prefix vector has finite support $0 \leq s < R_0 \leq Q$, while the terminal slack generating series is rational. For $p > q$, this boundary system closes the entire minimal-onset family. For $q > p$, the finite wall is added by the matrix inversion lemma; hence:

Theorem 8.1 (Complete minimal-onset algebraicity). *For every fixed coprime p, q , the generating series*

$$\mathcal{G}_{p,q}(t; x) = \sum_{r \geq 1} G_{p,q}^{(r)}(x) t^{r-1}$$

is algebraic over $\mathbb{Q}(x, t)$ and is given by a finite small-root/boundary system. No infinite-dimensional boundary problem remains.

For $q = 1$, $Q = 1$ and ordinary Lagrange inversion gives a scalar coefficient theorem. At 2:1 this simplifies further: if $W = 1 + xtW^3$, then

$$\boxed{\mathcal{G}_{2,1}(t; x) = x^4 W^3,}$$

so at the minimal r -carry scale $\kappa_r = r + 2$,

$$\boxed{G_{2,1}^{(r)}(x) = \frac{1}{r} \binom{3r}{r-1} x^{r+3}.}$$

9 Centered phase normal form and arbitrary-scale compression

Minimal onset still uses a distinguished reset skeleton. At arbitrary scale, BOTTOM and OUTER frames can be nested. The decisive fact is that the apparent binary-stack complexity collapses on the rational chamber.

For a stack word S , write

$$\rho(S) = qN_1(S) - pN_0(S), \quad R = -\rho(S) = pN_0(S) - qN_1(S).$$

The variable R is signed; on the rational chamber it is bounded below by $-hm$. The shifted counter $J = R + hm$ is therefore nonnegative. Let $A_{10}(S)$ and $A_{01}(S)$ denote the numbers of adjacent factors 10 and 01. A word is called *phase normal* if it consists of an alternating prefix followed by a monochromatic tail. The following theorem is proved by simultaneous induction on the physical transitions; the induction constructs two canonical reconstruction maps ν_0, ν_1 on the reachable shortest-demand pairs.

Theorem 9.1 (Centered phase-normal form). *For every configuration reachable from a rational-Dyck prefix:*

1. *the stack Parikh vector is the shortest-demand pair $\mu(R)$;*
2. *there is a phase bit $b \in \{0, 1\}$ such that*

$$\boxed{S = \nu_b(\mu(R));}$$

thus the complete bit stack is reconstructible from (R, b) ;

3. *the centered debt satisfies*

$$\boxed{A_{10}(S) - h \leq d \leq h - A_{01}(S).}$$

The three assertions are transition-closed only when proved simultaneously.

Proof. Define the two reconstruction maps explicitly by

$$\nu_0(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases} \quad \nu_1(x, y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

They have the same Parikh vector (x, y) and are exactly the two phase-normal orderings compatible with a shortest-demand pair. We prove simultaneously by induction over the displayed centered transitions that (i) the current Parikh vector is $\mu(R)$, (ii) the stack equals one of these two words, and (iii) the debt inequality holds.

For a BOTTOM-WRITE the exact shortest-demand formulas are

$$(x, y) = (d + q\omega, p\omega - d - 1) \quad \text{on input 1,} \quad (x, y) = (d + q\omega - 1, p\omega - d) \quad \text{on input 0,}$$

with the minimal positive weight ω of Theorem 6.2. Minimality of ω makes the minority coordinate at most h , so the canonical frame lies in the centered alternating interval and satisfies the debt bounds.

For CANCEL, deleting the visible matching cell decreases exactly one Parikh coordinate. Because the pre-transition pair is kernel-reduced and the deleted letter is visible in a phase-normal representative, the remaining pair is again the unique shortest pair for the updated signed demand; deleting the first letter of ν_b leaves a phase-normal word and changes A_{10}, A_{01} by at most one in the direction compensated by the unchanged debt.

For an interior SEARCH, the visible mismatching cell is consumed while d moves by one. If a SEARCH+ sees a stack beginning 00, phase normality forces the stack to be monochromatic from that point onward; hence $A_{01} = 0$ and the upper debt inequality remains strict enough after $d \mapsto d + 1$. The SEARCH- case beginning 11 is the physical dual. All alternating-prefix SEARCH cases simply shorten the alternating prefix and preserve both inequalities.

At OUTER+, the new frame has

$$X_+ = h + 1 + qc, \quad Y_+ = pc - h - 1,$$

with $Y_+ \leq h$ and a terminal zero-run, so it is one of the displayed ν_b words and resets d to 0 inside the required interval. OUTER- is the dual transition at the level of the induction; Theorem 9.2 below then excludes it on the rational-Dyck chamber. These cases exhaust the transition table, proving the three assertions simultaneously. The explicit reconstruction and compressed successor table are collected in Appendix A. $\square$

Theorem 9.2 (No negative outer carry). *OUTER- never occurs on a rational-Dyck prefix.*

Proof. At a hypothetical OUTER- trigger, $d = -h$ and the top symbol is 1. The lower centered inequality in Theorem 9.1 forces $A_{10}(S) = 0$. Phase normality therefore makes the entire nonempty stack 1^r . Hence the global rational height is

$$\rho_{\text{global}} = dm - \rho(S) = -hm - rq < 0,$$

contradicting rational-Dyck nonnegativity. $\square$

9.1 Exact one-counter recurrence

Theorem 9.1 makes the physical stack a derived object. The exact compressed state is

$$\sigma = (d, J, b), \quad J = R + hm,$$

where $d \in [-h, h]$, the signed demand residual $R = pN_0(S) - qN_1(S)$ satisfies $R \geq -hm$ on the rational chamber, hence $J \in \mathbb{Z}_{\geq 0}$ is a genuine one-counter coordinate, and b is the phase bit. When evaluating transitions we recover $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$; Appendix A gives the exact case table. If $\mu(R) = (x, y)$, popping a top 1 sends

$$R \mapsto R + q,$$

while popping a top 0 sends

$$R \mapsto R - p.$$

The next phase bit is uniquely determined by the reconstructed remaining shortest-demand word. OUTER+ sends

$$R' = (R - p) + (h + 1)m, \quad d' = 0,$$

and an empty-stack BOTTOM-WRITE on bit a sends

$$R' = dm + w(a), \quad d' = 0.$$

By Theorem 9.2, no OUTER− transition is present in the rational chamber.

For each positive-height compressed state $\sigma = (d, J, b)$, let $F_\sigma(z, x, y)$ be the generating series of factor-free accepting continuations, with x marking CANCEL, y marking OUTER+, and z marking WRITE scale weight. Appendix A gives, for each next bit a , an explicit deterministic successor $T_a(\sigma)$ and monomial $M(\sigma, a)$. Therefore

$$F_\sigma = \sum_{a \in \{0,1\}} M(\sigma, a) \Xi(T_a(\sigma)),$$

where Ξ returns 1 at the first legal empty-stack terminal state, 0 if the rational height becomes negative or a proper zero-residual contact occurs, and otherwise returns the corresponding successor series. The initial factor-free series $I(z; x, y)$ is obtained by the two possible first BOTTOM transitions, and

$$I_{k,r}(x) = [z^k y^r] I(z; x, y).$$

The only unbounded coordinate is the nonnegative integer J ; (d, b) is finite control. This is thus an exact weighted one-counter recurrence, to which standard one-counter/PDA generating-function methods apply [9, 1].

Corollary 9.3 (Arbitrary-scale operational enumeration). *For every coprime p, q and every scale k , the factor-free bivariate polynomial $I_k(x, y)$ is determined exactly by the compressed one-counter recurrence above. All rational-Dyck paths and operational Bizley cumulants follow from*

$$D(z; x, y) = \frac{1}{1 - I(z; x, y)}, \quad \log D(z; x, y) = \sum_{k \geq 1} \frac{A_k(x, y)}{k} z^k.$$

The primitive and minimal-onset formulas are specializations of the same compressed dynamics.

10 The original 2:1 family at arbitrary scale

The route returns to the motivating ratio with a particularly rigid all-scale law. In a factor-free scale- k 2:1 path, each WRITE has scale weight 1, every nonterminal WRITE episode contains exactly one SEARCH+, and the terminal episode contains none. If W, S, C are the numbers of WRITES, SEARCHes, and CANCELs, stack-height and input-length balance give

$$C + S = 2W, \quad W + C + S = 3k.$$

Hence $W = k$, $S = k - 1$, and

$$\boxed{C = k + 1}$$

for every factor-free path.

If an arbitrary rational-Dyck path of scale k has $s(P)$ zero-residual factors, additivity therefore gives the pathwise identity

$$\boxed{C(P) = k + s(P)}.$$

This is the operational contribution; the factor counts themselves belong to the classical Fuss–Catalan/factor-free theory.

The number of paths with s factors is

$$\boxed{\#\{P : C(P) = k + s\} = \frac{s}{k} \binom{3k - s - 1}{k - s}, \quad 1 \leq s \leq k,}$$

so

$$\boxed{D_k(x, 1) = \sum_{s=1}^k \frac{s}{k} \binom{3k - s - 1}{k - s} x^{k+s}.$$

The operational Bizley cumulants simplify to

$$\boxed{A_k(x, 1) = x^k \sum_{s=1}^k \binom{3k - s - 1}{k - s} x^s,}$$

and $A_k(1, 1) = \frac{1}{3} \binom{3k}{k}$ as required.

11 Computational validation and structural checks

The proofs above are symbolic. Exact computation was used independently as regression testing while the formulas were developed. The carry-onset hierarchy was checked on coprime $p, q \leq 7$ through scales up to 10 (350 scale tests and 138 onset tests for the first four carry layers) with no discrepancy. The stationary inter-carry determinant/Hessenberg formula was compared with direct local-word enumeration on 481 small parameter/slack pairs, and the rational Toeplitz bulk formula on 660 tested transitions, again with zero discrepancies. The scalar $p:1$ formulas were compared polynomial-by-polynomial with direct centered-machine enumeration in representative 2:1, 3:1, and 4:1 cases. Before Theorem 9.2 was proved, exact state-DP searches over 34 ordered coprime ratios with $p, q \leq 7$ and 176 ratio/scale instances found no accepted rational-Dyck example containing OUTER–; the theorem now explains that observation.

These computations are not used as proof. They serve to detect sign, parity, boundary, and reset errors in formulas whose symbolic proofs are independent.

12 Discussion

The event statistics exhibit a sharp structural transition. Primitive normalization has one fixed reference word; CANCEL is therefore a single Hamming/Schubert statistic and inherits matroid convexity. A physical WRITE reset replaces that static geometry by sums and products of Schubert pieces. For example, the first-carry polynomial at 5:3 is

$$36x^{10} + 67x^{11} + 153x^{12} + 180x^{13} + 135x^{14} + 39x^{15} + 18x^{16} + 2x^{17},$$

whose coefficients are not log-concave. Thus the loss of primitive Lorentzian behavior is not an artifact of proof technique; it records genuine history sensitivity introduced by a reset.

The repeated-kernel theory can be summarized as

physical WRITE budget + Raney carry skeleton + Schubert cancellation phases + Bizley zero-contact factorization
--

The phase-normal theorem then shows that this history-sensitive system is nevertheless much smaller than the literal binary stack suggests: on the rational chamber it is exactly a one-counter process with finite phase control.

13 Conclusion

Starting from the concrete 2:1 completion-aware DPDA, the residual viewpoint extends to a centered arbitrary-coprime $p:q$ normalizer and exposes a hierarchy of operational enumeration problems. Primitive CANCEL is a Hamming/Schubert statistic; repeated normalization creates a WRITE/carry hierarchy with Raney signatures and an algebraic stationary edge kernel; arbitrary scale collapses through phase normality to an exact one-counter recurrence. The special 2:1 family closes the circle with the pathwise law $C(P) = k + s(P)$.

The classical combinatorial theories used here remain classical. The new mathematical object is the event history of the centered normalization process and the exact identification of that history with those enumerative structures.

A Phase reconstruction and exact compressed transition table

For a shortest-demand pair (x, y) define the two phase-normal words

$$\nu_0(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases}$$

$$\nu_1(x, y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

When one of the formulas coincides with the other, the phase bit is immaterial. The simultaneous induction in Theorem 9.1 proves that every stack reachable from a rational-Dyck prefix is one of these two words for its shortest-demand pair.

It is important that the signed stack-demand residual need not itself be nonnegative. Put

$$R = pN_0(S) - qN_1(S) = -\rho(S).$$

The rational-chamber condition is

$$dm + R \geq 0.$$

Since $-h \leq d \leq h$, it follows that $R \geq -hm$. Thus the genuine one-counter coordinate is the shifted variable

$$J := R + hm \in \mathbb{Z}_{\geq 0}.$$

This shift corrects the misleading shorthand $R \geq 0$: R is signed, whereas J is the nonnegative counter.

Given a compressed state (d, J, b) , recover the signed value $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$ (with the empty-stack case treated separately by the BOTTOM rule). The exact transition and its weight monomial are:

condition	(d', R')	weight
$S = \zeta T, \zeta = a$	$(d, R + w_S(a))$	x
$S = 0T, a = 1, d < h$	$(d + 1, R - p)$	1
$S = 1T, a = 0, d > -h$	$(d - 1, R + q)$	1
$S = 0T, a = 1, d = h$	$(0, (R - p) + (h + 1)m)$	yz^c
$S = 1T, a = 0, d = -h$	excluded on the rational chamber	--
$S = \varepsilon$	$(0, dm + w(a))$	$z^{\omega_B(d, a)}$

where $w(1) = q$, $w(0) = -p$, $w_S(1) = q$, $w_S(0) = -p$ records the change in R under a pop, $c = 1 + \lfloor h/p \rfloor$, and ω_B is the exact bottom-WRITE weight from Theorem 6.2. In the first row, equivalently, popping a 1 sends $R \mapsto R + q$ and popping a 0 sends $R \mapsto R - p$. After computing the successor stack by the centered rule, b' is the unique phase for which $S' = \nu_{b'}(\mu(R'))$ (or is ignored when the stack is empty). This gives an executable deterministic recurrence on the finite control (d, b) and the single nonnegative counter $J = R + hm$; after each row one stores $J' = R' + hm$.

For a factor-free positive-height state, the corresponding generating series therefore satisfies

$$F_{d, J, b} = \sum_{a \in \{0, 1\}} M(d, J, b; a) \Xi(T_a(d, J, b)),$$

where M is the monomial in the table and Ξ is 1 on the first legal empty-stack terminal state, 0 after a negative rational height or a proper zero-residual contact, and otherwise the successor F -series. This is the precise one-counter recurrence used in Corollary 9.3.

B Carry-onset lower bound and wall-rank lemma

Let

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \quad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

Before the first positive outer carry, the maximal all-1 trajectory is the cheapest way to accumulate the required positive mismatch debt. It consumes exactly a units of scale before the first carry. After each OUTER+ reset, the written frame has length $cm - 1$, hence, by WRITE-weight conservation, each additional carry consumes at least one further block of scale weight c . Zeros cannot reduce either requirement because they decrease the rational height available to sustain the positive boundary trajectory. The all-1 extremal prefix followed by the unique shortest balancing continuation realizes equality at every level. Consequently the first scale supporting r carries is $\kappa_r = a + rc$.

At $k = \kappa_r$, a proper zero-residual factor would split the total scale into two positive balanced blocks. The carry-onset bound applied separately to the factors would then leave strictly less than the scale required for r carries in the factor containing them, contradicting minimality. Thus every r -carry path at first onset is factor-free.

For steep $q > p$, recall

$$\Theta = pQ - m(h + 1) > 0, \quad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

Then

$$b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor,$$

and because $m(h+1)/p > 1$ in every genuine wall case,

$$1 \leq b \leq Q - 1.$$

Hence every row needed by the finite Woodbury wall correction lies inside the already solved $Q \times Q$ boundary Green block. No additional start-state formula is required.

C Formal domain of the small-root formulas

The Q small roots of

$$v^Q = tK_c(v; x)$$

are naturally Puiseux series in an algebraic extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric combinations used for the Green functions descend to algebraic formal series over $\mathbb{Q}(x, t)$; the underlying coefficient recurrence has polynomial coefficients in x . Formulas containing x^{-Y} are therefore interpreted first over $\mathbb{Q}(x)$ and specialized at singular values such as $x = 0$ only through the polynomial recurrence or by a removable-limit argument. Likewise, Vandermonde-denominator expressions are interpreted for distinct generic roots and extended to collisions by symmetric continuation.

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